



Division of Chemistry and Chemical Engineering

Alex Siegel
1200 E. California Blvd., MC 147-75
Pasadena, CA 91125
(818) 439-7884
asiegel@caltech.edu

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SUBJECT: Cover Letter for Alex Siegel

Dear Calico job search committee,

I'm writing to apply for the position of Scientist in Protein Sciences at Calico. My career goal is to help unravel the molecular mechanisms underlying various age-related diseases brought on by protein misfolding, and to develop novel therapeutics that improve human health.

I received my Ph.D. in Biophysics with an additional specialization in Chemical Biology from the University of California, Berkeley in 2016. My thesis work on the mechanisms of bacterial transcription was carried out with Professor David E. Wemmer, specializing in protein NMR. I am currently a Postdoctoral Scholar in Chemistry studying molecular chaperones with Professor Shu-ou Shan at Caltech, using fluorescence spectroscopy, NMR, EPR, mass spectrometry, among other techniques in biochemistry and biophysics. Through several long-term collaborations, I gained further proficiency in protein NMR (Peter Wright, The Scripps Research Institute), cryo-electron microscopy (Yifan Cheng, UCSF), crosslinking-coupled mass spectrometry (Tsui-Fen Chou, Caltech; Feixia Chu, UNH), and plant biology (Bernhard Grimm, UH Berlin; Peng Wang HKU).

My recent postdoctoral work revealed how two different molecular chaperones recognize their clients and contribute to their proper folding. (1) We discovered a conformational switch in the chloroplast chaperone, cpSRP43, that allows it to function as a thermostat that rapidly transforms into a heat-protective chaperone following its partial unfolding at high temperature. This work improves our understanding of how essential photosynthetic complexes are delivered and maintained in response to heat stress. (2) We solved the cryo-EM structure of Skd3, a partial ClpB homolog found in human mitochondria. Surprisingly, Skd3 assembles into two complexes: a hexamer that disaggregates its clients and a unique dodecamer that forms a protective cage around clients to assist with their refolding. These projects led to three equal contribution first author papers in *Journal of Molecular Biology*, *Nature Plants*, and *Science Advances*, with a fourth in revision at *Science Advances*, and several other co-authorships in *Journal of Biological Chemistry* and *PNAS*. I am interested in further research on chaperones, protein folding, and especially on diseases related to protein aggregation. I hope my research on the detailed mechanisms of chaperones and misfolded proteins can contribute to the development of novel therapeutics to ameliorate the effects of aging and combat neurodegenerative diseases.

I have experience in all the experimental techniques necessary to carry out the specific responsibilities of this position at Calico. I regularly measure protein-ligand and protein-protein binding by NMR, anisotropy, ITC, FRET, among other methods. I analyze protein aggregation by light scattering and cross-linking coupled mass spectrometry, among other methods. I have over decade of experience with cloning, expression, chromatography purification, and labeling of proteins for biochemical and biophysical experiments.

Throughout my career, I have carried out independent research and developed new projects. My ideas and work have contributed to three funded grants, I've also published in high impact journals and presented at international conferences. I have actively sought to mentor many younger scientists and lead research projects in small teams. I directly trained 20 scientists, including: 1 postdoc, 7 PhD students, 6 undergraduates, and 6 high school students. I would be excited to work independently or lead small teams to tackle new research projects related to aging.

I greatly appreciate your time and consideration! Please don't hesitate to reach out to me if you have any questions or clarifications about my application.

Sincerely,

Alex Siegel
Postdoctoral Scholar, Chemistry